

RESEARCH EVALUATION REPORT

ResNet-50 Feature Extraction & Fuzzy Commitment Biometric Template Protection Scheme

Evaluated Datasets: Fingerprint_dataset (60 Subjects) & Multi-Impression FVC | Feature Extractor: ResNet-50 GAP (2048-dim)
| Crypto: SHA-256

1. Introduction & Objectives

This research report presents an empirical evaluation of a biometric template protection pipeline utilizing a deep ResNet-50 convolutional neural network (Global Average Pooling layer) for feature extraction and a Fuzzy Commitment scheme for cryptographic binding. The system converts raw fingerprint representations into fixed-length 2048-bit binary vectors, binds them to cryptographically secure random secrets, and evaluates secret recovery performance across single-capture verification, cross-capture multi-impersonation matching, and synthetic bit-flip robustness tests.

2. System Architecture & Cryptographic Hash Algorithm

Cryptographic Hash Algorithm: SHA-256 (Secure Hash Algorithm 256-bit, NIST FIPS 180-4 standard).

Feature Pipeline:

Fingerprint Image (224x224 RGB) $\longrightarrow$ ResNet-50 (GAP) $\longrightarrow$ $F \in \mathbb{R}^{2048}$ $\longrightarrow$ Binarization $\longrightarrow$ $B \in \{0, 1\}^{2048}$

Enrollment Phase:

- Secret Generation: Random secret key $S \in \{0, 1\}^K$ generated using a CSPRNG ($K \in \{64, 128, 256\}$).
- ECC Encoding: S is expanded via block repetition into codeword $C = \text{ECC_Encode}(S)$ of length $N = 2048$ bits (Repetition factor $R = 2048/K$).
- Helper Data: $W = B \oplus C$ (bitwise XOR).
- Cryptographic Commitment: Stored Hash $H = \text{SHA-256}(S)$ and Public Key $PK = g^S \bmod p$.

Authentication Phase (Verification 1:1):

- Query Presentation: Query biometric vector B_{auth} presented for claimed identity.
- Helper Data XOR: $C' = B_{\text{auth}} \oplus W = C \oplus (B_{\text{auth}} \oplus B_{\text{enroll}})$.
- ECC Majority Decoding: Candidate secret $S' = \text{ECC_Decode}(C')$ via majority voting over R -bit blocks.
- Hash Comparison: Candidate hash $H' = \text{SHA-256}(S')$. Verification checks exact string equality $H' == H_{\text{stored}}$.

How Hash Verification Works: Cryptographic hashes like SHA-256 do not use partial distance/threshold metrics. Due to the cryptographic avalanche effect, if the candidate secret S' is 100% identical to S (achieved via ECC error correction), $\text{SHA-256}(S')$ produces an exact 100% hash match against H_{stored} . If S' differs by even 1 bit, the hash check fails completely (0% match).

3. Individual Verification Output Table (Subjects 1 to 60)

Below is the complete verification table for all 60 subjects in Fingerprint_dataset (processed numerically 1 to 60). For each subject, the ResNet-50 GAP feature vector was extracted, enrolled, and tested against its stored helper data. All 60/60 subjects successfully recovered the exact secret key and matched the stored SHA-256 cryptographic commitment.

Subj #	Image Filename	Stored Commitment (SHA-256)	Candidate Hash (SHA-256)	Errors	Status
1	108_L0_0.bmp	2dbf74da518735e2368b...	2dbf74da518735e2368b...	0	MATCH
2	109_L0_0.bmp	a206a27a6f9f88afaa8b...	a206a27a6f9f88afaa8b...	0	MATCH
3	110_L0_0.bmp	55f7e0a805106f7a2db7...	55f7e0a805106f7a2db7...	0	MATCH
4	111_L0_0.bmp	79806edf31b921a45503...	79806edf31b921a45503...	0	MATCH
5	112_L0_0.bmp	d6921de3d0b8c759b0d0...	d6921de3d0b8c759b0d0...	0	MATCH
6	113_L0_0.bmp	ad1544073f3f325e874...	ad1544073f3f325e874...	0	MATCH
7	114_L0_0.bmp	c6eedef6c3725048cd35...	c6eedef6c3725048cd35...	0	MATCH
8	115_L0_0.bmp	d8aa9dd6a9301f292e38...	d8aa9dd6a9301f292e38...	0	MATCH
9	116_L0_0.bmp	00a53c6839c4c9b37b89...	00a53c6839c4c9b37b89...	0	MATCH
10	117_L0_0.bmp	5f66be2b02af460a50d7...	5f66be2b02af460a50d7...	0	MATCH
11	118_L0_0.bmp	9e61e4c70adf486e1b82...	9e61e4c70adf486e1b82...	0	MATCH
12	119_L0_0.bmp	576b1f33a168fedb1495...	576b1f33a168fedb1495...	0	MATCH
13	120_L0_0.bmp	bb19984927bd4a01c3d0...	bb19984927bd4a01c3d0...	0	MATCH
14	121_L0_0.bmp	e4e78ae00e43b231b443...	e4e78ae00e43b231b443...	0	MATCH
15	122_L0_0.bmp	08f250a1ac191d8bca50...	08f250a1ac191d8bca50...	0	MATCH
16	123_L0_0.bmp	d12866b0f656b8113187...	d12866b0f656b8113187...	0	MATCH
17	124_L0_0.bmp	690ff2089f79fa1f083a...	690ff2089f79fa1f083a...	0	MATCH
18	125_L0_0.bmp	dac63cd9e3315f241ba5...	dac63cd9e3315f241ba5...	0	MATCH
19	126_L0_0.bmp	a3d3d59dd5b6d275a23c...	a3d3d59dd5b6d275a23c...	0	MATCH
20	127_L0_0.bmp	7191a0fc47ee914f78a8...	7191a0fc47ee914f78a8...	0	MATCH
21	128_L0_0.bmp	eb382f3f7b396730d37f...	eb382f3f7b396730d37f...	0	MATCH
22	129_L0_0.bmp	b526e8801fe7c9032ac6...	b526e8801fe7c9032ac6...	0	MATCH
23	130_L0_0.bmp	efbdb518891669a04d2e...	efbdb518891669a04d2e...	0	MATCH
24	131_L0_0.bmp	2d8cb7774f3740dcbcdf...	2d8cb7774f3740dcbcdf...	0	MATCH
25	132_L0_0.bmp	d9e7a5caa26cf2dec39d...	d9e7a5caa26cf2dec39d...	0	MATCH

26	133_L0_0.bmp	65d9c95603832a5bd132...	65d9c95603832a5bd132...	0	MATCH
27	134_L0_0.bmp	5f1d46f146f5b440a2b5...	5f1d46f146f5b440a2b5...	0	MATCH
28	135_L0_0.bmp	40a162690f481571b6e0...	40a162690f481571b6e0...	0	MATCH
29	136_L0_0.bmp	b4c10699efa5ba1244c9...	b4c10699efa5ba1244c9...	0	MATCH
30	137_L0_0.bmp	e78095bc48bc7fb8774b...	e78095bc48bc7fb8774b...	0	MATCH
31	138_L0_0.bmp	452be36cb6874b45c63a...	452be36cb6874b45c63a...	0	MATCH
32	139_L0_0.bmp	49b43e7ee499f9656c11...	49b43e7ee499f9656c11...	0	MATCH
33	140_L0_0.bmp	ea631197182a7f3ecc11...	ea631197182a7f3ecc11...	0	MATCH
34	141_L0_0.bmp	4c0e122ecdc9cfc1f51b...	4c0e122ecdc9cfc1f51b...	0	MATCH
35	142_L0_0.bmp	3f1f0f5f7949c6f6aa87...	3f1f0f5f7949c6f6aa87...	0	MATCH
36	143_L0_0.bmp	e357a4db83e83291b113...	e357a4db83e83291b113...	0	MATCH
37	144_L0_0.bmp	ff4222fd089d6d2ad6f9...	ff4222fd089d6d2ad6f9...	0	MATCH
38	145_L0_0.bmp	34be7d8d97d751a25ee0...	34be7d8d97d751a25ee0...	0	MATCH
39	146_L0_0.bmp	a66bd815ea172269440f...	a66bd815ea172269440f...	0	MATCH
40	147_L0_0.bmp	fcda140b6a4b17778f5f...	fcda140b6a4b17778f5f...	0	MATCH
41	148_L0_0.bmp	2b0d4bad83c1db26ae0e...	2b0d4bad83c1db26ae0e...	0	MATCH
42	149_L0_0.bmp	3e6126807fb20a07741c...	3e6126807fb20a07741c...	0	MATCH
43	150_L0_0.bmp	f590c31feb99a7a6ebc2...	f590c31feb99a7a6ebc2...	0	MATCH
44	151_L0_0.bmp	7a69ee6799964d99389a...	7a69ee6799964d99389a...	0	MATCH
45	152_L0_0.bmp	6a9f2ba1335d367c6fe1...	6a9f2ba1335d367c6fe1...	0	MATCH
46	153_L0_0.bmp	e3e189429eb4850d6d9e...	e3e189429eb4850d6d9e...	0	MATCH
47	154_L0_0.bmp	802e7f5bf51c5389dbbd...	802e7f5bf51c5389dbbd...	0	MATCH
48	155_L0_0.bmp	4bc5b4fe0acfd9162aaf...	4bc5b4fe0acfd9162aaf...	0	MATCH
49	156_L0_0.bmp	ae97f5d467df1a1cc9c8...	ae97f5d467df1a1cc9c8...	0	MATCH
50	157_L0_0.bmp	22deb015eadf2a03871f...	22deb015eadf2a03871f...	0	MATCH
51	158_L0_0.bmp	23c5a81c140e3d7c9b82...	23c5a81c140e3d7c9b82...	0	MATCH
52	159_L0_0.bmp	bc9223226ff3a7e0c9c3...	bc9223226ff3a7e0c9c3...	0	MATCH
53	160_L0_0.bmp	019428de25c632b45241...	019428de25c632b45241...	0	MATCH
54	161_L0_0.bmp	f3aeb29a11d00314f9eb...	f3aeb29a11d00314f9eb...	0	MATCH
55	162_L0_0.bmp	c7cdf519fe736ea83e0f...	c7cdf519fe736ea83e0f...	0	MATCH

56	163_L0_0.bmp	464167a17c213a567da1...	464167a17c213a567da1...	0	MATCH
57	164_L0_0.bmp	d55fa1b3279292e987f2...	d55fa1b3279292e987f2...	0	MATCH
58	165_L0_0.bmp	22554f2749433416c433...	22554f2749433416c433...	0	MATCH
59	166_L0_0.bmp	4ed4dc6b5bac6cd4be67...	4ed4dc6b5bac6cd4be67...	0	MATCH
60	167_L0_0.bmp	3eddf2a17d49e9de3a53...	3eddf2a17d49e9de3a53...	0	MATCH

4. Binarization Strategy & Parameter Comparison

Two binarization strategies were evaluated across all 60 subjects (3,540 imposter trials):

- Median Binarization: Per-sample median thresholding.
- Standardized Binarization: Channel-wise zero-mean whitening fitted strictly on enrollment features (zero test data leakage).

Binarization Method	Secret Size (K)	Rep Factor	Imp. False Acceptances	FAR (%)	FRR (%)	EER (%)
Standardized	256 bits	8x	0 / 3,540	0.0000%	0.00%	0.00%
Standardized	128 bits	16x	46 / 3,540	1.2994%	0.00%	0.65%
Standardized	64 bits	32x	689 / 3,540	19.4633%	0.00%	9.73%
Median	256 bits	8x	132 / 3,540	3.7288%	0.00%	1.86%
Median	128 bits	16x	243 / 3,540	6.8644%	0.00%	3.43%
Median	64 bits	32x	1,734 / 3,540	48.9831%	0.00%	24.49%

5. Cross-Capture Multi-Impression Evaluation

To evaluate real intra-subject biometric variability, an experiment was conducted using multiple distinct captures of the same finger (10 subjects x 8 impressions = 80 images). Impression 1 was enrolled to create helper data W, and Impressions 2 through 8 were presented for authentication.

Key Findings:

- Average Intra-Subject Hamming Distance: Different captures of the same finger differed by an average of 567.9 bits out of 2048 (27.7% Bit Error Rate).
- Cross-Capture Genuine Secret Recovery: With Median Binarization (K=128 bits, 16x repetition factor, maximum block error tolerance = 7 bits), the scheme successfully recovered the secret across 37.14% of cross-impression genuine attempts without explicit image registration or minutiae alignment.

6. Synthetic Bit-Flip Robustness Experiment

To evaluate the theoretical error correction bounds of the Block ECC decoder, controlled synthetic bit-flip noise was injected into query binary vectors across all 60 subjects:

Synthetic Noise (%)	Expected Bit Errors (N=2048)	Successful Verifications	Recovery Rate (%)
2.0%	~41 bits	60 / 60	100.00%
5.0%	~102 bits	60 / 60	100.00%
10.0%	~205 bits	60 / 60	100.00%
15.0%	~307 bits	55 / 60	91.67%
20.0%	~410 bits	34 / 60	56.67%

7. Conclusions & Defensible Claims

Based on the empirical evidence gathered across single-capture, cross-capture, and synthetic noise evaluations:

1. The experimental results indicate that the current implementation achieves 100% genuine secret recovery on the evaluated single-capture dataset.
2. The synthetic noise experiment indicates successful secret recovery up to 10% bit-flip noise for the evaluated subjects.
3. Channel standardization (whitening) combined with K=256 secret bits achieves FAR = 0.00% across 3,540 imposter trials on the evaluated dataset.
4. Raw biometric templates are not stored in plaintext. Further security analysis is required to evaluate irreversibility and resistance to template reconstruction attacks.